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The Impact of "Lazy Minting" on Seller Performance in NFT Marketplaces—A Transaction Cost Economics Perspective

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ABSTRACT

In the burgeoning marketplaces of digital assets, non-fungible tokens (NFTs) revolutionize digital asset ownership and intellectual property (IP) protection, but high minting costs create barriers to marketplace entry and growth. This study examines the impact of "lazy minting," a new NFT production method introduced by major NFT marketplaces to lower minting costs by deferring blockchain certification until the first sale. In response to the call for further research on emerging technologies in operations management, we explore how this policy affects the net sales performance of existing sellers in the NFT marketplaces. Based on transaction cost economics (TCE) and the literature about different IP protection methods, we distinguish between lazy- and regular-minted NFTs by their differential transaction costs and utilize the staggered difference-in-differences (DID) method to conduct our analysis. We find that lazy minting adoption significantly boosts the net sales performance of existing sellers. This is attributed to their cost-adaptive IP protection behavior. Specifically, they achieve this by minting more NFTs with a larger proportion of style-consistent NFTs through lazy minting, while strategically employing regular minting for style-breaking NFTs, which is contingent upon their reputation. Our study has important theoretical and practical implications for operations management under the emerging technological revolution.

1 | Introduction

Global investors are increasingly drawn to non-fungible tokens (NFTs),[1] which, as unique blockchain entries, verify digital asset ownership and authenticity (Pun et al. 2021; Umar et al. 2022), offering intellectual property (IP) protection for items like art and media (Bamakan et al. 2022; Bonnet and Teuteberg 2023b). NFT marketplaces, exemplified by OpenSea with its 3 million users and $4.5 million daily trading volume,[2] enable NFT discovery, listing, and trading, fostering a vibrant enthusiast community. NFT minting, crucial for digital asset creation in marketplaces, involves uploading assets to the blockchain and ensuring immutable, verifiable ownership.[3] Despite its IP protection benefits, the regular minting method poses high gas fees

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Summary

- Lazy minting, as a novel NFT minting way, reduces the ex ante transaction costs through eliminating the minting gas fees, but increases the ex post transaction costs from the diminished IP protection strength.
- Lazy minting adoption boosts the net sales performance of existing sellers in the NFT marketplaces.
- Existing sellers achieve better sales through strategically minting more style-consistent NFTs by lazy minting, while still employing regular minting for style-breaking NFTs.
- High-reputation sellers are more daring to create style-breaking NFTs, even when utilizing the lazy minting method.

(ranging from a few dollars to tens or even hundreds of dollars,[4] depending on network congestion and file size[5]) that may outweigh a significant portion of the sales revenues, which are often below $300.[6] As over one-third of OpenSea sales fall below $100, gas fees account for a substantial portion of the sales income.

More recently, major NFT marketplaces, such as OpenSea and Rarible, have introduced "lazy minting." This method enables NFT creation without initial gas fees, deferring blockchain certification until the first sale. Lazy-minted NFTs are listed off-chain, merging minting and transferring ownership into a single transaction upon purchase. Therefore, only the gas fees associated with the combined on-chain transaction need to be paid. In contrast, for regular-minted NFTs, creators must pay the minting gas fees in the minting process regardless of the potential sales in the future. Nevertheless, the adoption of lazy minting may introduce new problems. The zero cost of lazy minting reduces the entry barrier to the NFT market, which risks a flood of NFTs entering the market.[7] This free minting feature also has weak IP protection as it is not certified via blockchain. NFT sellers therefore face a trade-off between the two minting methods: Lazy minting saves costs but lacks an immediate blockchain record to prove ownership,[8] risking duplication and suggesting *weak IP protection strength*,[9] while regular minting, though costlier, confirms a clear establishment of ownership by immediate blockchain record, indicating *strong IP protection strength*.

Given the trade-off between lazy minting and regular minting, the existing sellers with a history of transactions in the NFT marketplaces must carefully consider and strategize their minting operations in view of the new option of lazy minting. However, we know very little about whether and how its adoption has implications for the existing sellers' net sales performance (i.e., revenue minus transaction and minting gas fees if any) in the NFT marketplaces, given the nascent nature of the phenomenon. This leads to our research question: "To what extent and how does lazy minting adoption by existing sellers affect their NFT net sales performance in NFT marketplaces?"

To answer this research question, we contemplate the essence of the trade-off between the two minting methods by drawing on transaction cost economics (TCE) (Williamson 1985). Minting costs, low for lazy minting and high for regular minting, epitomize a typical *ex ante cost* in TCE. IP protection enforced through regular minting to prevent infringement and reduce the cost of dispute resolution represents a way to reduce *ex post cost* in TCE (Benaroch et al. 2016). Thus, the existing sellers' adoption of lazy minting, as a means to reduce *ex ante cost* per TCE, entails sellers mitigating higher *ex post cost* resulting from infringement risks due to the nature of lazy minting. To this end, we reason that the style signature of the primary content of NFTs, that is, artworks, is an important factor for sellers to consider in conjunction with their lazy minting adoption decision due to its implication to infringement risk mitigation. Specifically, *style-consistent artworks* (i.e., the artworks that follow a consistent style of an artist) require less protection due to higher market recognition and lower infringement risks, whereas *style-breaking artworks* (i.e., the artworks that break a consistent style of an artist) demand more due to lower initial recognition resulting from its newness (Ackerman 1962). In addition, IP protection demand due to infringement risks may also vary depending on the artist's reputation, which contributes to the market recognition of artworks and shifts the level of uncertainty (Atanasov et al. 2012; Davison 2010; Halaburda et al. 2023; Li et al. 2013; Standifird and Weinstein 2007). Thus, we employ the TCE and consider the distinctiveness of artwork style signatures (style-consistent vs. style-breaking) and artist reputation to formulate our hypotheses.

We conduct an empirical analysis based on the implementation of lazy minting in OpenSea using the staggered difference-in-differences (DID) method (Beck et al. 2010) and conduct a series of robustness checks including the Callaway and Sant'Anna difference-in-differences method (CSDID) (Callaway and Sant'Anna 2021). We first perform analyses from the successful transaction perspectives to examine the effect of lazy minting adoption on existing sellers' net sales performance. Then, we carry out mechanism analyses to explore IP protection behavior changes—specifically, minting behavior reflected by minting frequency and minting method utilization (regular or lazy minting based on style signature)—and the moderating effect of artist reputation. In addition, we conduct a series of additional analyses to examine corresponding changes in the supply side and demand side, respectively.

Our investigation reveals that the adoption of lazy minting by existing sellers catalyzes increased transaction volumes, leading to enhanced net sales performance. This effect is primarily due to their cost-adaptive IP protection behavior, which is manifested in the production of more NFTs, with a higher proportion of style-consistent NFTs created by lazy minting and the reserved use of regular minting for style-breaking NFTs. The moderating influence of the sellers' reputations further supports our arguments, indicating that sellers with higher reputations are more inclined to utilize lazy minting with increased audacity, even for NFTs exhibiting lower style consistency. Additional results from both supply and demand perspectives further demonstrate the cost-adaptive IP protection behavior.

This research makes several theoretical contributions. First, it provides insights into how lazy minting adoption affects the net sales performance of existing sellers, contributing to NFT marketplace studies and responding to calls for research on digital transformation (Angelopoulos et al. 2023) and advancing technology implementation (Heim et al. 2021) in operations management. Second, we respond to the call for TCE extensions to the novel phenomenon driven by technological advances (Cuypers et al. 2021). By linking ex ante and ex post costs with the minting method, NFT style signature, and seller reputation, our article extends TCE's applicability to the blockchain-enabled NFT marketplaces and the domains of IP protection for digital creative artworks. Third, it enriches the literature on artwork style signatures (Gohoungodji and Amara 2023; Stamkou et al. 2018), especially through the analytical definition of style consistency using AI algorithms. Fourth, it adds to the body of knowledge on IP protection in NFTs (Bamakan et al. 2022; Bonnet and Teuteberg 2023b) by illustrating the strategic use of minting methods with different IP protection strengths in the NFT marketplaces, echoing the call for a better understanding of blockchain technology in operations management (Babich and Hilary 2020; Zhao et al. 2022). This study also provides practical guidance to NFT sellers on how to strategically mint their NFT works to yield better net sales performance, while providing managers with valuable reference on the adoption and implementation of lazy minting.

2 | Literature Review

2.1 | Overview of Transaction Cost Economics (TCE)

Transaction Cost Economics (TCE) originated from the work of Ronald Coase in 1937 and was further developed by Oliver Williamson, focusing on the costs associated with economic exchanges and the governance structures that arise to minimize these costs (Williamson 1975). Over the past decades, TCE has been widely applied to supply chain management (Cuypers et al. 2021; Grover and Malhotra 2003), inter- and intra-firm transactions (Jiang et al. 2022), knowledge management (Sharma et al. 2020; Skowronski et al. 2020; Wagner and Bode 2014), and consumer actions (Akturk et al. 2018; Griffis et al. 2012; Son et al. 2020).

According to TCE, balancing ex ante and ex post costs is key to minimizing transaction costs, achieving greater efficiency, and reducing risks associated with unforeseen complications (Bhardwaj and Ketokivi 2021; Grover and Malhotra 2003). Ex ante costs, incurred before finalizing a transaction, fall into two main categories: drafting and negotiating costs, as well as safeguard costs (Benaroch et al. 2016; Ketokivi and Mahoney 2020; Zhang et al. 2021). Drafting and negotiating costs encompass resources spent on creating detailed contracts and reaching mutually agreeable terms, which can be significant when complex contingencies are involved. Safeguard costs involve measures to protect parties' interests, such as establishing credible commitments or common ownership to mitigate transaction-related risks. In contrast, ex post costs arise after transaction completion, primarily focusing on enforcement. These include expenses for monitoring contractual compliance and resolving disputes, which can significantly impact overall transaction efficiency (Baldia 2013; Ketokivi and Mahoney 2020). Due to the interdependence of ex ante and ex post costs, it is essential to address them simultaneously rather than in a sequential manner (Williamson 1985). The recommended approach is to adopt a forward-looking perspective that anticipates potential issues—such as contracts with strong protection strength—rather than merely responding to problems after they arise (Ketokivi and Mahoney 2020). For example, Benaroch et al. (2016) examine how contract design choices in software development outsourcing can achieve a balance between ex ante and ex post transaction costs to optimize overall transaction cost management.

One of the important factors when choosing different governance structures to balance the ex ante and ex post costs and minimize the total transaction costs is uncertainty (Fan et al. 2024; Sharma et al. 2020). Uncertainty can be classified into environmental uncertainty, which pertains to unpredictable changes in the external environment, and behavioral uncertainty, which refers to the difficulty in predicting how parties involved in a transaction will act in unforeseen circumstances that are not explicitly covered by a contract (Cuypers et al. 2021). This type of uncertainty arises from the inherent unpredictability of human behavior, where one party not only struggles to anticipate the actions of the other party but also finds it challenging to predict their own responses in unprecedented situations (Grover and Malhotra 2003; Ketokivi and Mahoney 2020). A substantial body of literature investigates the various factors influencing uncertainty (Fan et al. 2024; Fan et al. 2022) and the adaptive strategies employed by firms in response. For example, Oxley (1999) finds that U.S. firms engaging in international R&D alliances tend to employ more hierarchical governance mechanisms to mitigate uncertainty, particularly when partnering with firms from countries that have weaker patent protections.

Apart from the determining factors like uncertainty, some social mechanisms can act as a "shift parameter" that affects the relative transaction costs (Williamson 1991). A "shifting parameter" can alter the cost–benefit analysis of different governance modes (e.g., market, hierarchy, or hybrid forms) and can lead to shifts in the decision-making process regarding how transactions are organized. Taking trust level as an example, variations in societal trust can affect the willingness of parties to engage in market transactions versus hierarchical governance. For instance, in societies with high trust, firms may prefer market transactions, while in low-trust environments; they may opt for more hierarchical arrangements to mitigate risks (Oxley 1999; Williamson 1991). Apart from that, reputation serves as a crucial shifting parameter within TCE, impacting transaction costs and governance structures. It refers to the perceived reliability and trustworthiness of transacting parties, which can significantly influence their willingness to engage in transactions (Halaburda et al. 2023). A strong reputation can not only reduce uncertainty and the likelihood of opportunistic behavior of the suppliers (Standifird and Weinstein 2007; Williamson 1991), but also mitigate the risk of infringement faced by suppliers because of a higher probability of being held liable (Atanasov et al. 2012; Li et al. 2013), thereby lowering transaction costs.

In OM literature, existing research mainly utilizes TCE to study supply chain management (Johnson et al. 2007; Power and Singh 2007; Wagner and Bode 2014) and outsourcing decisions (Fan et al. 2022; Holcomb and Hitt 2007). However, the rapid development of various technologies (Angelopoulos et al. 2023; Heim et al. 2021) has the potential to influence both the level of behavioral uncertainty that the firms face and the relative costs of different governance mechanisms (Cuypers et al. 2021). Despite this, the impact of emerging technologies on the relative costs across different mechanisms and subsequent operational decisions remains underexplored. Cuypers et al. (2021) specifically highlight the potential of blockchain technology to reduce uncertainty and opportunism, thereby lowering transaction costs. In response, Halaburda et al. (2023) propose a theoretical perspective on smart contracts as a novel governance structure. However, there is currently a lack of empirical evidence regarding the impact of blockchain-based applications on operational decisions from a transaction cost perspective.

2.2 | Transaction Costs in IP Protection of Digital Creative Works

In the creative industry, IP protection aims to prevent unauthorized use and reproduction of artworks, safeguarding artists from financial losses and ensuring that they maintain control over their creations (Zorloni 2013). Since creativity is a scarce resource in the industry, IP protection enables artists to capitalize on the economic value of their unique creations, allowing them to command high prices in the market, thereby encouraging continued production and innovation (Benghozi and Santagata 2001; Zorloni 2013). Furthermore, effective IP protection is crucial for the growth and sustainability of the industry, as it encourages investment and supports the economic activities associated with creative works (Throsby 1994).

However, traditional IP protection methods and their underlying legal frameworks were originally predicated on tangible assets that possess exclusivity (Patry 2011). When it comes to safeguarding digital creative works, which can be reproduced and disseminated widely (Dwivedi et al. 2022), conventional IP protection methods such as copyright, patents, and trademarks present numerous challenges (Barrère and Delabruyère 2011; Bonnet and Teuteberg 2023a; Mishra et al. 2024). On the one hand, the high ex ante costs associated with conventional IP protection methods are ill-suited to the high-frequency nature of digital art creation (Mishra et al. 2024). For example, due to the rigorous standards of novelty and nonobviousness requirements for design patents and trademarks, obtaining protection is a time-consuming process that involves significant acquisition costs (Bonnet and Teuteberg 2023a; Hrytsai 2023; Wang et al. 2024). Even the process of registering a copyright typically takes several months (Bamakan et al. 2022). Moreover, IP protection through traditional methods often incurs high search costs due to a lack of formalities (Bonnet and Teuteberg 2023a). This can lead to the problem of "orphan works" (Lee 2023), which are copyrighted works whose owners are unknown or cannot be located. Additionally, the transfer of IP under traditional methods generally necessitates duly signed written contracts, which involve high drafting and negotiation costs (Hrytsai 2023). On the other hand, the widespread dissemination of digital creative works diminishes the enforcement efficacy of traditional IP protection methods. Due to the disparities in protective measures across different countries and regions (Bonnet and Teuteberg 2023a), IP protection certifications based on one country's or region's standards often struggle to gain recognition in others (Forero-Pineda 2006; Helpman 1992).

The limitations of conventional IP protection methods in safeguarding digital creative artworks have given rise to Digital Rights Management (DRM) (Bechtold 2004). DRM is a set of technologies and policies designed to control the access, use, and distribution of copyrighted digital content (Finck and Moscon 2019). Traditional DRM systems restrict access to digital content based on licensing agreements. Users must obtain a license to access or use the content, and the DRM system enforces these restrictions. In contrast to conventional IP protection methods, DRM requires ongoing costs. For instance, EZDRM Multi-DRM pricing amounts to $299.99 per month.[10] Due to its high cost, it is only utilized by a small group of people, like large corporations with thousands of IPs (Fairfield 2022a). Nevertheless, the encryption methods employed by DRM are often easily circumvented, leading to content leakage (Fairfield 2022a; Mishra et al. 2024). Moreover, the subsequent enforcement costs associated with DRM pose significant challenges. Not only do different countries vary in their recognition and acceptance of DRM, but the independent operation of various DRM platforms also gives rise to copyright disputes and mutual non-recognition issues (Koenen et al. 2004; Melendez-Juarbe 2009).

The limitations inherent in conventional IP protection methods and DRM systems underscore the imperative for innovative solutions in the realm of digital creative artwork protection. NFTs, with their distinct characteristics of indivisibility and provenance, based on blockchain technology (Patrickson 2021), have redefined IP protection (Bonnet and Teuteberg 2023b; Lee 2023). NFTs establish unique and exclusive ownership of digital creative artworks, addressing their historical devaluation due to fungibility (Fairfield 2022a; Lee 2023). Unlike physical masterpieces such as the Mona Lisa, digital artworks previously lacked discernible originals, hindering their marketability. NFTs solve this issue by creating unique, blockchain-based tokens representing digital art, thereby establishing scarcity and authenticity in the digital realm (Bamakan et al. 2022). This approach allows for unfettered propagation and appreciation of the digital asset, enhancing its value as it circulates freely online (Zeilinger 2018).

NFTs can reduce both ex ante and ex post costs compared to traditional IP protection methods. First, NFTs harness blockchain's efficiency to significantly reduce the time and cost associated with traditional IP protection methods. By providing immediate, verifiable ownership, NFTs offer a compelling solution while awaiting formal patent or trademark approval (Fairfield 2022b). Second, NFT represents a system of IP protection that directly serves individual artists, thus avoiding the searching and negotiation costs due to intermediaries like major copyright industries, distributors, middlemen, and gatekeepers (Kraizberg 2023; Lee 2023). Furthermore, NFT reduces the enforcement costs. The immutable records maintained on blockchain provide reliable proof for exclusive IP (Lee 2023; Wang et al. 2024). These tamper-resistant entries can be presented as evidence in a court of law (Hrytsai 2023).[11] Additionally, blockchain's transparency and immutability enable NFT transactions to transcend geographical and platform limitations, addressing mutual non-recognition issues in traditional IP protection and DRM systems (Bamakan et al. 2022). These features collectively make NFTs a cost-effective solution for the IP protection of digital creative artworks.

TABLE 1 | Transaction costs for different IP protection methods of digital creative works.

IP protection ways	Ex ante costs	Ex post costs
Patent/Trademark/ Copyright	High	High
DRM	Medium	High
NFT (Regular minting)	Medium	Low
NFT (Lazy minting)	None	High

However, the situation differs between lazy-minted and regular-minted NFTs. While neither requires contract drafting or negotiation in NFT marketplaces, lazy minting, unlike regular minting, avoids the minting gas fees. This reduces costs for creators, especially during network congestion. However, this cost reduction comes at the expense of diminished IP protection strength in comparison to regular minting. The underlying reason for this trade-off is that lazy-minted NFTs are not stored on the blockchain but on the platform's centralized server until their initial sale (Brennan 2022; Ko et al. 2024). Without immutable blockchain records as proof, ownership of the digital artworks becomes controversial when infringement occurs, thereby incurring higher enforcement costs. Tian (2023) posits that lazy minting may potentially lead to creators losing control over their works. Moreover, lazy-minted NFT metadata is vulnerable to manipulation, potentially leading to unfair trading (Yan et al. 2024). Table 1 summarizes the transaction cost comparison between different IP protection methods of digital creative works.

While the potential of NFTs in IP protection and their various practical applications have been extensively discussed (Bonnet and Teuteberg 2023b; Lee 2023), there remains a dearth of attention on their actual usage. Recent scholarly attention has primarily focused on pricing and value dynamics within these NFT marketplaces (Hofstetter et al. 2024; Horky et al. 2022; Yang 2024). The distinctive operational characteristics of NFT marketplaces and their influence on user behavior have also garnered scholarly interest. For example, Tunç et al. (2021) investigate the impact of compulsory resale royalties in NFT marketplaces, a feature not typically found in traditional IP protection methods.

Our study aims to contribute to the literature by examining the strategic use of two distinct NFT minting methods with varying IP protection strengths in NFT marketplaces, focusing on lazy minting, which lowers ex ante costs. Existing literature has suggested that similar policies could increase market thickness, lead to higher search friction, and lower overall matching efficiency (Geva et al. 2019; Li and Netessine 2020), with evidence found in NFT marketplaces by Zhang et al. (2022). However, the impact of such policies on existing sellers' behavior and net sales performance remains unexplored. While reduced market-level efficiency might adversely affect existing sellers, the discretionary nature of minting choices allows for strategic utilization of lazy minting, potentially shielding sellers from negative market-level repercussions. Considering the pivotal role of existing sellers' sales performance in two-sided markets (Cheng et al. 2023), our study thus zeroes in on those existing sellers, exploring how and through what mechanisms the lazy minting adoption affects net sales performance. From the perspectives of both artworks and artists, we incorporate two factors that potentially influence net sales performance by affecting the level of uncertainty and, consequently, transaction costs.

2.2.1 | Artwork Characteristics—Style Consistency of Digital Creative Works

According to TCE, one of the most important factors that minimize transaction costs (i.e., determine the strategic utilization of different IP protection methods) is uncertainty (Fan et al. 2024; Grover and Malhotra 2003; Skowronski et al. 2020). One potential factor highly correlated with the level of uncertainty for digital creative artworks is their style signature (Ackerman 1962). *Artist style signature* refers to the distinguishable and unique set of characteristics in an artist's work that identifies it as their own. It encompasses the consistent elements of form, technique, and expression that collectively convey the artist's personal vision and approach to art. The artists' artworks can accordingly be classified into two types: *style-consistent artworks* and *style-breaking artworks* (Ackerman 1962; Simmel 1991; Stamkou et al. 2018). Style-consistent artworks adhere to the established style signature of an artist or the prevailing style of a particular era or movement. They reflect continuity in artistic expression and align with the expected norms of style (Ackerman 1962; Simmel 1991). Style-breaking artworks are artworks that deviate from an artist's signature style or the predominant style of their contemporaries. These pieces often challenge the status quo and lead to a new direction in the artist's works or influence the evolution of art as a whole (Simmel 1991; Stamkou et al. 2018).

Both style-consistent and style-breaking artworks are important for artists, representing different aspects of artistic development and public reception. Style-consistent artworks reinforce the artist's brand and satisfy audience expectations, providing a sense of reliability and trust in the artist's vision (Chan 1992; Person and Snelders 2010). On the other hand, style-breaking artworks demonstrate the artist's versatility, willingness to innovate, and ability to contribute to the discourse of art. They can generate new interest and dialogue, potentially leading to a reevaluation of the artists' work and an increase in their influence within the art world (Sgourev and Althuizen 2014; Stamkou et al. 2018). The balance between these two types of artworks is crucial for an artist's growth and the dynamic nature of his oeuvre.

Artworks with a consistent style, which have well-established authenticity (Adler 2023), are often perceived as having a lower risk of plagiarism. Their recognizable aesthetic, which can be easily attributed to a specific artist, deters potential plagiarists (Ackerman 1962). Copying such works would be more easily detectable and could be claimed as trademark infringement due to the developed secondary meaning associating the style with the artist (Mendelson-Shwartz et al. 2023). Thus, from the perspective of TCE, *the behavioral uncertainty is low for style-consistent artworks*. For instance, consider Andy Warhol's infringement of Patricia Caulfield's photograph in his "Flowers" series. Caulfield's photograph, adhering to a distinct style easily associated with her, became the subject of a legal dispute when

Warhol used it without consent. The lawsuit was quickly settled in Caulfield's favor,[12] demonstrating the lower uncertainty of style-consistent works.

However, style-breaking artworks, by their very nature, challenge established norms and introduce novel elements that can not immediately be attributed to a specific creator or tradition. This inherent ambiguity in origin and ownership can elevate the risk of plagiarism, as it becomes more challenging to discern and prove the originality of such works. Consequently, *the behavioral uncertainty, including those risks associated with legal enforcement and dispute resolution, is higher for style-breaking artworks*. Ackerman's theory of style suggests that the initiation of a style makes it difficult to determine who the innovator is (Ackerman 1962). This obscurity is particularly pronounced in style-breaking artworks, where the departure from conventional aesthetics can blur the lines of authorship. When an artwork does not conform to a recognizable style, it becomes more susceptible to multiple ownership claims. This can potentially lead to legal disputes over IP rights. These disputes can result in significant financial losses, not only due to the costs of litigation but also due to the potential devaluation of the artwork when its authenticity or originality is questioned. As the innovative effort put into it is usually high, the need for strong IP protection to mitigate the uncertainty is heightened for style-breaking artworks.

2.2.2 | Seller Characteristics—Artists' Reputation

According to TCE, reputation is a key shifting parameter of governance mechanism decisions (Halaburda et al. 2023), which reduces the possibility of behavior uncertainty (Atanasov et al. 2012; Li et al. 2013; Standifird and Weinstein 2007). In the art world, an artist's reputation is the public and professional recognition of their creative work and influence, which can enhance their artwork's market recognition (Lang and Lang 1988). A well-established reputation serves as a multifaceted asset in the context of IP protection. First, it enhances the likelihood of obtaining robust legal support, as entities with high reputations often possess greater resources and credibility within the legal system. This increased access to legal recourse acts as a deterrent against potential infringers (Davison 2010). Second, the prominence associated with a high reputation inherently reduces the probability of IP infringement, as the visibility and recognition of the sellers' work make unauthorized use easier to detect. Lastly, the strong market position accompanying a high reputation implies that sellers are more inclined to maintain their standing, thereby reducing the likelihood of opportunistic behavior. This, in turn, diminishes buyers' uncertainty and transaction costs, facilitating their purchase decisions (Standifird and Weinstein 2007).

3 | Hypothesis Development

In our scenario, following the implementation of a lazy minting policy, sellers are at liberty to choose between lazy minting and regular minting. From the perspective of TCE, the extent of formal controls should be a function of uncertainty (Handley and Benton Jr. 2013; Williamson 1985). We propose that ideally, after lazy minting adoption, sellers should employ "cost-adaptive IP protection," that is, strategically utilizing these two minting methods to balance ex ante costs with ex post costs depending on NFT style signature, thereby minimizing transactional costs, enhancing productivity and ultimately maximizing net sales performance (Bidwell 2010; Schulze and Zellweger 2021). Below, we analyze the two available minting options for sellers (lazy minting vs. regular minting) and the two styles of works (style-consistent vs. style-breaking) from the standpoints of IP protection and transaction costs, and further propose our hypotheses.

Specifically, we argue that utilizing lazy minting for style-consistent artworks and regular minting for style-breaking artworks can balance ex ante and ex post transaction costs and thereby optimize overall net sales performance (Benaroch et al. 2016). Lazy minting, with its lower ex ante costs, is an effective function to create more NFTs without heavy cost burdens. To alleviate the risk caused by the lack of blockchain copyright validation, style-consistent artworks are suitable for utilizing the lazy minting method, as they inherently carry a lower uncertainty due to their recognizable aesthetic. Moreover, given the higher market recognition for this segment of NFTs, it could lead to an overall enhanced net sales performance for the existing sellers (Bidwell 2010). Regular minting, despite higher ex ante costs for safeguarding, affords stronger IP protection that is beneficial for style-breaking artworks, which are of higher uncertainty due to their innovative nature. This approach mitigates the potential ex post costs of legal disputes over IP rights.

Therefore, we predict that after sellers adopt lazy minting and realize its nature of cost and IP protection strength, they will utilize this method to mint more style-consistent artworks and thereby achieve higher sales performance and reserve the utilization of regular minting for their style-breaking artworks. By aligning the minting method with the artwork's uncertainty profile (i.e., cost-adaptive IP protection behavior), creators can reduce the total transaction costs—both ex ante and ex post—enhancing the overall productivity and net sales performance of their creative outputs (Bidwell 2010). Therefore, we propose the following hypotheses:

H1. *Existing sellers who adopt lazy minting achieve higher net sales performance (i.e., revenue minus transaction and minting gas fees if any) compared with sellers who do not adopt it.*

H2. *(Mechanism: cost-adaptive IP protection behavior): Existing sellers who adopt lazy minting will mint more NFTs with a higher proportion of style-consistent NFTs by lazy minting and reserved utilization of regular minting for style-breaking NFTs.*

In online marketplaces, sellers who have engaged in a substantial number of transactions are often perceived as more reputable (Deng et al. 2023; Ye et al. 2009). This correlation can be attributed to the cumulative effect of transactional history, where each successful exchange enhances the seller's credibility

(Standifird and Weinstein 2007). Consequently, a robust transactional record serves as a proxy for reliability, fostering trust among potential buyers and contributing to the seller's standing within the marketplace. Particularly in the context of NFT marketplaces, the absence of a review and rating system renders past transactions a more precise and reliable metric of reputation compared to other scenarios. This is because the blockchain's immutable record provides a transparent and verifiable track of a seller's transaction history, which, in the absence of traditional feedback mechanisms, becomes a critical indicator of reputation in the digital marketplace.

As we have discussed earlier, reputation reduces the level of behavioral uncertainty for both buyers and sellers (Atanasov et al. 2012; Halaburda et al. 2023; Li et al. 2013; Standifird and Weinstein 2007). Similarly, in the actual art trading market, most art forgers opt to imitate lesser-known, second-tier artists[13] because infringement is difficult to claim when the artist is not well known (Mendelson-Shwartz et al. 2023). However, for well-known artists, this dynamic shifts. Their high reputation not only deters potential infringers but also increases the likelihood of successful legal action against those who do infringe, thereby increasing the potential infringer's likelihood of being held legally accountable (Davison 2010). Moreover, a well-established reputation can expedite the transaction process by reducing the uncertainty of buyers (Schulze and Zellweger 2021). Consequently, for sellers with high reputations, even when employing lazy minting for their digital creative artworks, the accelerated successful transaction speed can ensure that the NFT is swiftly sold and subsequently recorded on the blockchain, thereby mitigating the risk of IP infringement. Therefore, we conjecture that existing sellers with higher reputation (i.e., more previous transactions) will be more inclined to use lazy minting, even for style-breaking artworks. The hypothesis H3 is as follows.

H3. *Existing sellers with more previous transactions are more likely to employ lazy minting for their NFTs with comparatively low style consistency than existing sellers with fewer previous transactions.*

4 | Research Context and Design

4.1 | Research Context

Our empirical setting is OpenSea, the first and largest two-sided marketplace for NFTs, which was launched in 2017. OpenSea has over one million users in 2024 and over 80 million different types of NFTs. As of early 2022, OpenSea broke the record of $2 billion in daily trading volume.[14] Initially, all the NFTs were stored on the blockchain at the minting stage, where their authenticity and ownership were established, with the creators paying the minting gas fees. On December 29, 2020, OpenSea launched lazy minting to allow creators to mint NFTs without any upfront gas costs, as they are not written on-chain until the first purchase or transfer is made.[15] Since the implementation of the lazy minting policy, creators have been able to choose which minting method to use when creating NFTs in OpenSea.

Nevertheless, there is no apparent label to separate lazy-minted NFTs from regular-minted NFTs, as all items are displayed in the same list on the platform. Buyers can only distinguish the minting method of each NFT based on whether it has a minting record on the Ethereum blockchain. OpenSea was the first NFT marketplace to launch the lazy minting function and the first to attempt unbundling the on-chain issuance of NFTs from the metadata. Before launching the lazy minting policy, OpenSea did not release any preview of this new function. After about 10 months, Rarible was the second marketplace to implement this policy on October 19, 2021.[16]

To illustrate the policy impact at the market level, Figure 1 depicts two key temporal metrics: the total number of NFTs minted across the entire market and the first-month sales rate of these newly minted NFTs. The left figure shows the number of NFTs minted using different methods. Following the introduction of lazy minting, the number of lazy-minted NFTs gradually increased, eventually dominating the market. However, as shown in the right figure, the sales rate of lazy-minted NFTs is significantly lower than that of regular-minted NFTs. This disparity led to a decline in the overall NFT sales rate, dropping from approximately

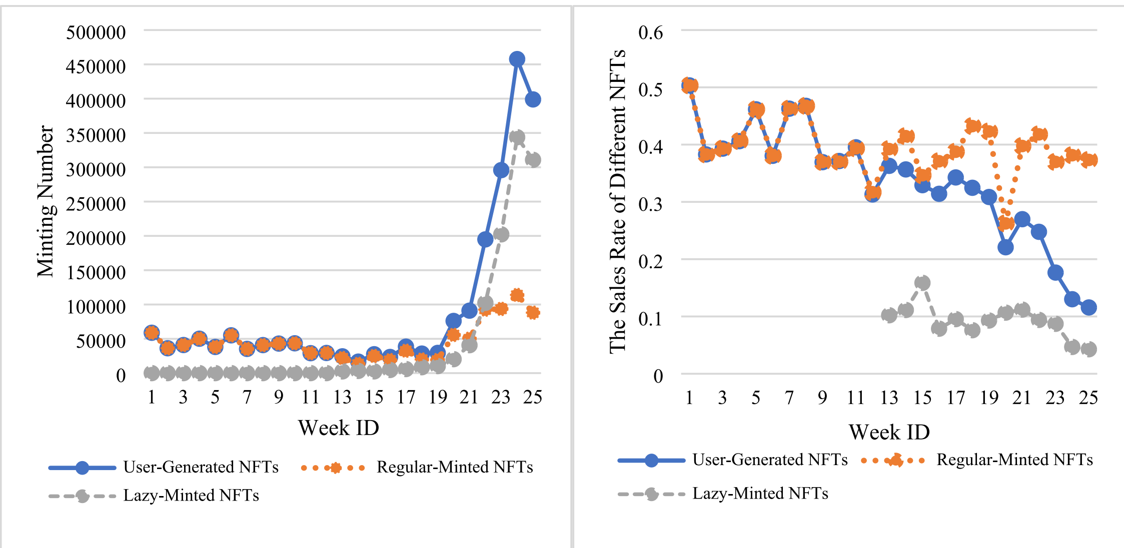

FIGURE 1 | Model-free illustration of NFT market trends: Minting number and sales rate.

40% before the introduction of lazy minting to nearly 10% within 3 months of its launch. This market-level phenomenon is due to the influx of new entrants, which gets lots of coverage. To be specific, lazy minting lowers the entry barrier to creating NFTs (no fees for NFT minting), so many newcomers flood into the market and create some plagiarized work (such as minting a picture downloaded from the internet).[17] OpenSea platform also confirmed this issue: "Over 80% of the items created with this tool were plagiarized works, fake collections, and spam."[18]

In NFT marketplaces, all NFTs can be classified into two categories: user-generated NFTs and third-party-generated NFTs. The minters of user-generated NFTs are also the creators of the digital works, so these minters have complete control over the design of their NFTs' content. They are more akin to creators of digital art, such as the famous NFT creator Mr. Doodle, a 23-year-old British artist. He attained international success thanks to his lively interlocking designs, and he announced the release of his NFTs in 2021. On the other hand, third-party-generated NFTs are designed by third-party companies, with minters minting the NFTs on the blockchain according to the addresses of corresponding smart contracts offered by the companies, such as the famous Bored Ape NFTs created by the Bored Ape Yacht Club. In this case, minters are not involved in the creation, so they cannot determine the NFTs' content design. Instead, they select the NFT collections created by third-party companies and then complete the subsequent minting steps. Our analysis mainly focuses on user-generated NFTs because their content can be designed by minters and represents the minters' creative and behavioral changes. We therefore identify the user-generated NFTs by their contract addresses and track their minting and transaction records from OpenSea and Ethereum data aggregator (i.e., dune.com).

4.2 | Data

Our data consists of all the users whose NFTs had successful transaction records before the lazy minting implementation in OpenSea. As OpenSea launched the lazy minting policy on December 29, 2020, we choose 3 months before and after this event as our observation period for the main analysis.[19] Specifically, we take the week of December 29, 2020, as Week 0 (December 28, 2020, to January 3, 2021) and define the 12 weeks before the launch of lazy minting (from October 5, 2020, to December 27, 2020) as the pre-event period, while the 12 weeks after the launch (from January 4, 2021, to March 28, 2021) are defined as the post-event period. According to the OpenSea blog,[20] no other key functions are updated in the observation period.

To observe changes in minting behavior and subsequent net sales performance of existing sellers, we select users who minted user-generated NFTs in both pre- and post-event periods. Users who adopted lazy minting in the post-event period are designated as the treatment group, while those who exclusively used regular minting throughout both periods form the control group. We collect minting records and relative images for all user-generated NFTs created by the sample users during the observation period. Subsequently, we track the transaction, auction, and offering records for these NFTs until the end of 2022 (2 years after the introduction of lazy minting) from OpenSea. The final sample comprises 446 users in the treatment group and 904 in the control group.

4.3 | Research Design

Considering the sellers' different starting enforcement times of the lazy minting policy, we use the staggered difference-in-differences method (Beck et al. 2010) to identify the policy's effect on existing sellers and utilize the CSDID method with the same datasets for robustness check (Callaway and Sant'Anna 2021). In our research scenario, users independently determine whether to adopt lazy minting. To mitigate self-selection bias, we utilize propensity score matching (PSM) to construct comparable treatment and control groups in our main analysis (Rosenbaum and Rubin 1983) and also use different matching covariates, coarsened exact matching, and lookahead matching to show robustness (details provided in Appendix G). For PSM, we utilize the covariates including the total transaction amount, transaction number, and minting number 12 weeks before the implementation of lazy minting. We perform a one-to-one nearest-neighbor match with a maximum matching radius of 0.001 for the treatment and control groups.[21] Table 2 shows the balancing test results before and after propensity score matching. After PSM, the biases of all the covariates between the treatment and control groups are far less than 10%, and there are 300 users in the treatment group and 344 users in the control group.

4.4 | Model Specification

Our model for the empirical analysis is:

$$Y_{it} = \beta_0 + \beta_1\, Treat_i \times Post_{it} + \beta_2\, Control_{it} + u_i + \tau_t + \epsilon_{it} \quad (1)$$

TABLE 2 | Result of propensity score matching.

	Before PSM				After PSM			
Variables	**Treat**	**Control**	**Bias%**	***t*-stat.**	**Treat**	**Control**	**Bias%**	***t*-stat.**
LogMintNum	2.594	2.152	41.4	36.25***	2.433	2.434	−0.1	−0.06
LogTransNum	2.042	1.778	19.3	17.06***	1.923	1.933	−0.7	−0.52
LogTransAmt	5.563	5.449	3.5	2.98***	5.412	5.395	0.5	0.38

Note: $^{***}p<0.01$; $^{**}p<0.05$; $^{*}p<0.1$.

where Y_{it} denotes the dependent variables including the variables about transaction behaviors and minting strategies. The definitions and summary statistics of all the dependent variables are shown in Tables 3 and 4. The variable $Treat_i$ is a dummy that equals one if seller i had lazy minting records, and zero if seller i only used regular minting. $Post_{it}$, which denotes when seller i started to use the lazy minting function, equals zero if seller i never used lazy minting before week t and one if seller i already had lazy minting records in week t. $Control_{it}$ represents a set of control variables for controlling the potential underlying trends, which will be defined in detail in corresponding sections. u_i captures the time-invariant user-fixed effects while τ_t captures time-fixed effects, and e_{it} captures any idiosyncratic random errors. With these specifications, the coefficient β_1 serves as our estimator, which captures the impact of adopting lazy minting on existing sellers' minting behavior and net sales performance.

TABLE 3 | Definitions of dependent variables.

Variables	Definitions
Successful transaction analysis	
TransNum	The number of successful transactions in the primary market within 3 months for NFTs minted by seller i in week t
AvgTransPrice	The average adjusted price of all transactions in the primary market within 3 months for NFTs minted by seller i in week t (amount in USD, and eliminate the cost of minting gas fees for regular-minted NFTs)
TransAmt	The transaction amount (net sales) in the primary market within 3 months for NFTs minted by seller i in week t (amount in USD, and eliminate the cost of minting gas fees for regular-minted NFTs)
Minting behavior analysis	
UserGenNFTNum	The number of user-generated NFTs minted by seller i during week t
RegularMintNFTNum	The number of user-generated NFTs minted with the regular minting method by seller i during week t
Visual Consistency	The average of the minimum spatial distances between the visual feature vectors of NFTs minted by seller i in week t and those minted previously
Content Consistency	The average of the minimum spatial distances between the content feature vectors of NFTs minted by seller i in week t and those minted previously

5 | Empirical Analysis

5.1 | Net Sales Performance Change of Existing Sellers

While the policy of lazy minting may introduce negative effects at the market level, such as reduced matching success rates and efficiency, we hypothesize that it offers existing sellers the potential to strategically utilize lazy minting to decrease overall transaction costs, ultimately leading to increased net sales performance. Therefore, we conduct an analysis of the post-adoption trading performance of existing sellers who have adopted lazy minting, utilizing the transaction number, the average transaction price, and the transaction amount as dependent variables, and accounting for the minting gas cost. For each transaction, we deduct the minting gas fees paid by the seller from the transaction price.[22] To control for potential underlying transaction trends, we include variables such as the total number of transactions, average transaction price, and transaction amount for NFTs minted by seller i in the month prior to week t. Following Flannery and Rangan (2006)'s method of dummy variable adjustment, we use a control dummy, *TransControlDum*, to solve the problem that some control variables in the panel are empty since not all sellers have minting records for every month. All the dependent variables satisfy the parallel trend assumption.[23]

According to the regression results shown in Table 5, lazy minting adoption has a positive and statistically significant effect on the transaction number (14.68%) of existing sellers. However, existing sellers utilizing lazy minting sell 5.26% fewer regular-minted NFTs after adopting lazy minting. Besides, they have a 78.96% higher transaction amount after the adoption, although the effect of lazy minting adoption on their NFTs' average transaction price is not statistically significant. These results indicate that existing sellers who adopt lazy minting get better net sales performance and the sales increment originated from the lazy-minted segment. We also add some other NFT-related time-varying control variables (as Table B1 in Appendix B shows), and the results still hold. Our hypothesis H1 is therefore supported.

Furthermore, we conduct additional analyses on the sales rate and sales waiting duration of existing sellers. As demonstrated in Appendix B, the sales rate for existing sellers who adopted lazy minting remains consistent with their pre-adoption levels (Table B2). However, the lazy-minted NFTs generated by these sellers exhibit a significantly longer waiting duration before a successful sale (Table B3). This finding indicates that while the sales rate of lazy-minted NFTs is not diminished for existing sellers, these NFTs require an extended period before they are sold and verified on the blockchain to establish ownership. This

TABLE 4 | Summary statistics of dependent variables.

Variables	Obs#	Mean	Std.	Min	Max
All user-generated NFTs					
LogTransNum	16,100	0.232	0.610	0	6.004
LogAvgTransPrice	2600	5.118	2.102	−3.911	13.171
LogTransAmt	16,100	1.010	2.400	−3.215	13.171
LogUserGenNFTNum	16,100	0.420	0.714	0	5.814
VisualConsistency	4972	22.250	7.462	0	46.267
ContentConsistency	4740	0.896	0.277	0	1.487
Only regular-minted NFTs					
LogTransNum	16,100	0.197	0.556	0	6.004
LogAvgTransPrice	2226	5.168	2.172	−3.911	13.171
LogTransAmt	16,100	0.879	2.267	−3.215	13.171
LogRegularMintNFTNum	16,100	0.349	0.632	0	5.024
VisualConsistency	4262	22.434	7.397	0	46.267
ContentConsistency	4028	0.908	0.271	0	1.479

Note: To measure net sales performance, we subtract the minting costs (gas fees for regular-minted NFTs) from the NFT's selling price, and we delete few outliers where gas fees largely exceed the selling price.

TABLE 5 | Successful transaction analysis.

Variables	*LogTransNum*	*LogAvgTransPrice*	*LogTransAmt*
NFT types		**All user-generated NFTs**	
Treat×Post	0.137*** (0.019)	−0.071 (0.101)	0.582*** (0.075)
TransControlDum	0.156*** (0.040)	0.213 (0.201)	0.433*** (0.103)
LogLastMonthTransNum	0.232*** (0.014)	0.048 (0.034)	—
LogLastMonthAvgTransPrice	−0.004 (0.005)	0.034 (0.026)	—
LogLastMonthTransAmt	—	—	0.191*** (0.021)
Constant	−0.029 (0.041)	4.890*** (0.194)	0.239** (0.108)
Observations	16,100	2600	16,100
R-squared	0.322	0.807	0.273
NFT types		**Only regular-minted NFTs**	
Treat×Post	−0.054*** (0.015)	0.066 (0.123)	−0.232*** (0.062)
TransControlDum	0.135*** (0.040)	0.071 (0.226)	0.261*** (0.107)
LogLastMonthTransNum	0.212*** (0.014)	0.039 (0.041)	—
LogLastMonthAvgTransPrice	−0.004 (0.005)	0.019 (0.029)	—
LogLastMonthTransAmt	—	—	0.159*** (0.021)
Constant	−0.001 (0.040)	5.050*** (0.220)	0.424*** (0.112)
Observations	16,100	2226	16,100
R-squared	0.314	0.822	0.272

Note: User- and week-fixed effects are included; robust standard errors are in parentheses. $^{***}p<0.01$; $^{**}p<0.05$; $^{*}p<0.1$.

prolonged verification delay underscores the weaker IP protection afforded to lazy-minted NFTs.

5.2 | Minting Behavioral Change of Existing Sellers: Cost-Adaptive IP Protection Behavior

5.2.1 | The Impact on the Minting Frequency

To investigate how the existing sellers would gain better net sales performance, which means to what extent and how existing sellers adopt lazy minting, we conduct the minting behavior analysis. First, we explore the direct impact of adopting lazy minting on existing sellers' minting frequency of different kinds of NFTs, with the numbers of user-generated NFTs and user-generated NFTs with the regular minting method as the dependent variables. Table 6 reports the regression results, with each of the two columns representing NFTs minted with different methods by these sellers. As the data shows, β_1 is consistently significant under the specifications in columns (1) and (2), suggesting that the existing sellers who adopt lazy minting mint substantially more (47.99%) user-generated NFTs than those who do not adopt lazy minting. However, those who adopt lazy minting are less willing (−13.76%) to use regular minting to create user-generated NFTs. In conclusion, the adoption of the lazy minting policy increases existing sellers' frequency of minting user-generated NFTs by lazy minting, while reducing their frequency of regular minting utilization.

5.2.2 | The Impact on the Minting Strategy

We further analyze the minting strategy change of existing sellers who adopt lazy minting. To differentiate the style-consistent and style-breaking NFTs and test our hypotheses, we construct the measure of the style consistency of each NFT based on two different features extracted by machine learning algorithms (Chou et al. 2023): visual feature and content feature. Visual feature is characterized by the combination and interaction of elements such as color, contour, texture, and brightness, which capture the visual perception of appearance (Attneave 1954; Shin et al. 2020). The content feature describes the objects and their relationship in the NFT, which captures the semantics and the content of the work (Shin et al. 2020). For visual features, we utilize the self-supervised visual representation learning algorithm DINOv2 to describe the pixel-level traits of each NFT through a 768-dimensional vector (Oquab et al. 2023; Zhou and Lee 2023). Next, for content features, we first utilize the multimodal model BLIP-2 to identify the objects of the NFT and transform the image or gif into a descriptive text (Li et al. 2023; Zhou and Lee 2023). Then we use the pre-trained text embedding model based on BERT to describe the text content in terms of a 384-dimensional spatial vector (Reimers and Gurevych 2019).

The style consistency of the NFT represents to what extent such NFT shares the same style with that seller's prior works. To measure the consistency of these two kinds of features, we conduct the Euclidean distance to measure the relative features of objects in a vector space, which is commonly used to measure the creativity of the image or identify the machine-made works (Boden 1998; Shin et al. 2020; Zhou and Lee 2023). For details, it is measured as the minimum spatial distance between the NFT and the prior NFTs created by that existing seller. In this way, we obtain two measurements of style consistency as the dependent variables: *visual consistency* and *content consistency*. The summary statistics for the visual consistency and content consistency at the NFT level are provided in Appendix C. The lower value of the two variables represents higher similarity to previous works and thus higher style consistency. Taking the NFTs of specific existing sellers' for instance, the three pictures in Figure 2 are all from the same seller *0xDeadthing*, who is famed for depicting figures with abstract lines and colors.[24] After converting them into vectors and calculating the consistency, we get a lower distance between the first/second one and the prior works, which means higher content consistency. For the third one, the distance is higher, indicating low content consistency.

We calculate the average consistency of the NFTs minted in a specific week by each seller and examine the impact of lazy minting adoption on style consistency. Because the measurement of consistency may be affected by the total number of NFTs previously minted by the seller, we control for the cumulative number of the users' previous minting. The results are shown in Table 7. The visual and content consistency scores of the existing sellers' user-generated NFTs significantly decrease after the lazy minting adoption; however, the style consistency scores of their regular-minted NFTs do not have a significant change after the adoption. It means that the existing sellers utilize the lazy minting method to mint NFTs with high style consistency to their prior NFTs, while utilizing the regular minting method to mint the style-breaking artworks (at the same innovation level as previous artworks before the adoption). To mitigate the potential issue of existing rare or atypical artworks misrepresenting a seller's style signature, we also utilize the 5% and 10% lowest percentile of the

TABLE 6 | Existing sellers' minting number change.

Variables	*LogUserGenNFTNum*	*LogRegularMintNFTNum*
Treat × *Post*	0.392*** (0.024)	−0.148*** (0.016)
Constant	0.365 (0.005)	0.370*** (0.005)
Observations	16,100	16,100
R-squared	0.265	0.258

Note: User- and week-level fixed effects are included; robust standard errors are in parentheses. ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

FIGURE 2 | Style-consistent NFTs and style-breaking NFTs.

TABLE 7 | Style consistency change of existing sellers.

Variables	*Visual consistency*	*Content consistency*	*Visual consistency*	*Content consistency*
NFT types	**All user-generated NFTs**		**Only regular-minted NFTs**	
Treat×Post	−1.149*** (0.402)	−0.053*** (0.016)	−0.556 (0.447)	−0.022 (0.018)
CumulativeMintNum	−0.020*** (0.003)	−0.001*** (0.000)	−0.011** (0.005)	−0.001*** (0.000)
Constant	23.055*** (0.162)	0.949*** (0.007)	22.736*** (0.156)	0.941*** (0.006)
Observations	4972	4740	4262	4028
R-squared	0.399	0.359	0.406	0.380

Note: User- and week-fixed effects are included; robust standard errors are in parentheses. ***$p<0.01$; **$p<0.05$; *$p<0.1$.

cosine distances between the focal NFT and the prior works of each seller as the measurements of visual consistency and content consistency. The results are shown in Table C2 and Table C3 of Appendix C, which can indicate the robustness of our findings.

Therefore, we can identify the existing sellers' minting behavior change after the lazy minting adoption. To balance the ex ante and ex post costs, the existing sellers mainly utilize lazy minting to create more style-consistent NFTs. Given the lower uncertainty associated with these style-consistent NFTs, the subsequent ex post costs are lower, necessitating a lesser degree of IP protection strength. This is reflected in the diminished style consistency score across the existing sellers' overall NFT portfolio. Meanwhile, they reserve the utilization of regular minting because of its strong IP protection for their style-breaking NFTs that carry a higher risk of being copied and thus incur greater ex post costs, that is, trading higher ex ante costs for reduced ex post costs. Due to the lower production costs and the sellers' proficiency in creating style-consistent NFTs, their total output increased as Table 6 suggests. In summary, H2 is well supported.

5.2.3 | The Moderating Effect of Existing Sellers' Reputation on IP Protection Behavior

Reputation, which is a set of attributes inferred from past actions of the individual or corporate (Weigelt and Camerer 1988), can effectively reduce the plagiarism risk of sellers, shift the level of uncertainty, and therefore reduce transaction costs (Davison 2010). We posit that the reputation of existing sellers plays an important role in moderating the market recognition of their NFTs and affects the degree to which different minting methods are used, as H3 stated.

To test our hypothesis, we partition the existing sellers into two sets based on the median number of transactions 1 year before the official launch of the lazy minting policy. If a seller's number of transactions is above the median of all sellers, then the seller falls into the high-reputation set. Otherwise, the seller falls into the low-reputation set. We repeat the staggered DID estimation on style consistency analyses of both groups. The findings, as delineated in Table 8, reveal that the diminution in style consistency score among NFTs minted by sellers of high reputation is significantly less pronounced than that observed among their low-reputation counterparts. The findings suggest that existing sellers with high reputations are more audacious in minting their NFTs, opting for lazy minting even when the similarity to their previous NFTs is not as pronounced. In contrast, sellers with a lower reputation exercise greater caution, favoring lazy minting for creations that closely resemble their prior works. These observations lend credence to our H3. To demonstrate the robustness of our results, we also employed the past transactions as moderating variables to test the moderating effect of reputation. Moreover, we utilize the transaction amount 1 year before the policy and the personal website disclosure of existing sellers as the group criteria to conduct the analysis. As shown in Appendix D, all the outcomes remain consistent with our findings.

TABLE 8 | Sellers' prior reputation to the style consistency.

Groups	High-reputation sellers		Low-reputation sellers	
Variables	*Visual consistency*	*Content consistency*	*Visual consistency*	*Content consistency*
NFT types	All user-generated NFTs		All user-generated NFTs	
Treat×Post	−0.746 (0.488)	−0.041** (0.019)	−1.936*** (0.716)	−0.075** (0.029)
CumulativeMintNum	−0.018*** (0.004)	−0.001*** (0.000)	−0.025*** (0.005)	−0.002*** (0.000)
Constant	22.728*** (0.211)	0.928*** (0.009)	23.692*** (0.273)	0.987*** (0.012)
Observations	3189	3075	1783	1665
R-squared	0.379	0.346	0.434	0.425
NFT types	Only regular-minted NFTs		Only regular-minted NFTs	
Treat×Post	−0.300 (0.542)	−0.009 (0.021)	−0.927 (0.799)	−0.033 (0.034)
CumulativeMintNum	−0.009* (0.005)	−0.001*** (0.000)	−0.034** (0.013)	−0.003*** (0.001)
Constant	22.333*** (0.198)	0.918*** (0.007)	23.694*** (0.254)	0.997*** (0.011)
Observations	2835	2721	1427	1307
R-squared	0.392	0.353	0.436	0.428

Note: User- and week-fixed effects are included; robust standard errors are in parentheses. ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

From the results above, we can confirm how the existing sellers strategically utilize lazy minting to decrease overall transaction costs without being affected by the low IP protection of the lazy minting method (i.e., cost-adaptive IP protection behavior). To analyze how such cost-adaptive IP protection behavior leads to sales increment, we employ the three-step mediation analysis method (Baron and Kenny 1986). Specifically, we incorporate the minting frequency (i.e., minting number) and the minting strategy (i.e., visual consistency and content consistency) as the mediating variables. As the results in Appendix E, the sales increment effect gets weakened after considering these mediation variables, which support the mediation path of sellers' cost-adaptive IP protection behavior to sales increment.

TABLE 9 | Behavior analysis of buyer engagement.

Variables	*The number of views*	
NFT types	All user-generated NFTs	Only regular-minted NFTs
Treat×Post	9.845*** (2.900)	−0.681 (3.127)
Constant	44.612*** (0.865)	45.967*** (0.791)
Observations	11,165	10,169
R-squared	0.372	0.391

Note: User- and week-fixed effects are included; robust standard errors are in parentheses. ***$p < 0.01$; **$p < 0.05$; *$p < 0.1$.

6 | Additional Analysis

6.1 | Additional Evidence for Sellers' Cost-Adaptive IP Protection Behavior

To corroborate the cost-adaptive IP protection behavior exhibited by existing sellers, we delve into an analysis of the post-event changes in the existing sellers' auction behavior, and the shifts in buyers' offering with the NFTs minted after the event, which is shown in Appendix F. The results indicate that, owing to the cost-saving benefits of lazy minting, sellers tend to set lower initial prices on average, affording themselves greater flexibility in price adjustments. Concurrently, the consistent style of lazy-minted NFTs ensures that buyers' perceived value remains undiminished. These findings further corroborate the mechanism of existing sellers' cost-adaptive IP protection behavior.

Then, we provide further evidence of cost-adaptive IP protection behavior from the perspective of changes in buyer engagement. If the existing sellers increasingly employ lazy minting for style-consistent NFTs, these NFTs, already enjoying higher market recognition, should elicit greater engagement from buyers in the market, as evidenced by metrics such as the frequency of views. We thus utilize the average number of views up to the end of 2023 for the NFTs minted by the existing sellers each week as the dependent variable and employ the same empirical model in the main analysis to conduct this analysis. The results in Table 9 indicate that the existing sellers' user-generated NFTs gain extremely higher views after the lazy minting adoption. However, the average views for their regular-minted segments do not have significant change. It shows that the large number of view increments comes from their lazy-minted NFTs, which means the existing sellers' lazy-minted NFTs have higher views and market recognition compared with their regular-minted NFTs, aligned with the cost-adaptive IP protection behavior.

6.2 | Alternative Explanation Regarding the Buyers' Discernment

To ensure that buyers who express interest in purchasing lazy-minted NFTs are experienced buyers with extraordinary abilities to assess the value of NFTs and rationally offer appropriate prices, rather than inexperienced buyers lacking knowledge of the NFT market, who are irrationally choosing lazy-minted

TABLE 10 | Buyers' discernment analysis (before events).

	Sales rate analysis			*Yield rate analysis*		
Buyer types	**Obs#**	**Average**	***t*-stat.**	**Obs#**	**Average**	***t*-stat.**
Non-Lazy Minting Buyers	13,146	0.158	1.855*	3128	1.245	−4.517***
Lazy Minting Buyers	2282	0.144		642	1.66	

Note: ***$p<0.01$; **$p<0.05$; *$p<0.1$.

TABLE 11 | Buyers' discernment analysis (DID).

	SalesRate	***LogYieldRate***
Treat × Post	0.018* (0.010)	0.176 (0.189)
Constant	0.209*** (0.002)	1.040*** (0.047)
Observations	34,053	9904
R-squared	0.519	0.478

Note: User- and week-fixed effects are included; robust standard errors are in parentheses. ***$p<0.01$; **$p<0.05$; *$p<0.1$.

NFTs and offering inappropriate prices. We partition the buyers into two sets based on whether they buy lazy-minted NFTs after the launch of lazy minting. Moreover, the analysis of the differences in their sales rate and yield rate is predicated on the following considerations. Buyers with higher discernment ability in trading NFTs can evaluate the value of NFTs more accurately and comprehensively, and they are more likely to obtain higher investment returns in NFT trading. We trace all purchases made by these two groups of buyers before and after the launch, as well as the resale of all kinds of NFT products they purchased.

We select the sales rate (i.e., the proportion of NFTs bought by buyer i in week t that were successfully sold by the end of 2022) and the yield rate (i.e., the ratio of the selling price to the buying price of NFTs bought by buyer i in week t that were successfully sold by the end of 2022) to evaluate the performance of buyers investing in NFTs. We first compare the sales rate and yield rate of these two groups of buyers before the launch of lazy minting, and the results are shown in Table 10. Our results indicate that buyers who engage in transactions related to lazy-minted NFTs have higher profits as shown in the yield rate but slightly lower resale rates before the policy launch. We then compare the sales rate and yield rate of buyers of lazy-minted NFTs and buyers of non-lazy-minted NFTs using the DID method, and the results are shown in Table 11. We find that the lazy-minted NFT buyers do not have worse sales performance, but instead get better sales rates. The results further support that our main findings are not the outcomes of the irrational offerings by inexperienced buyers.

7 | Robustness Checks

To confirm our empirical results, we conduct a series of robustness checks for our empirical analysis. Due to space constraints, in this section, we present a summarized overview of the robustness checks, while the complete result tables can be found in the Supporting Information. The robustness check analyses are as follows:

7.1 | Parallel Trends Analysis

First, we check the validity of our control group by analyzing the weekly parallel trends, as shown in Appendix A. We can confirm that prior to the adoption of lazy minting, the trends of transactions of existing sellers' NFTs, the minting behaviors of the existing sellers, and the engagement of the buyers are similar between the treatment group and the control group. Therefore, all the dependent variables in our analyses satisfy the parallel trend assumptions.

7.2 | Different Matching Methods

7.2.1 | Adding Covariates for PSM

To further avoid the self-selection bias, we add some user-level covariates for PSM, such as whether the user sets a name, whether the user provides a social media account, whether the user sets a banner image, whether the user has a personal website, and the proportion of prior works in GIF format. After the matching, there are 258 users in the treatment group and 291 users in the control group, with no significant difference in covariates between the two groups. We conduct the analysis of their transactions and minting behavior, and Table G1 of Appendix G shows the regression results, which are consistent with our findings.

7.2.2 | Coarsened Exact Matching (CEM)

Similarly, we also conduct the CEM method with the same covariates in the last section. After the matching, while there is a notable decrease in the degree of covariate imbalance, the total user sample of the control group and the treatment group is less than 200 people, which is too small to represent the user population. Therefore, we only conduct the successful transaction analysis for robustness, which is shown in Table G2 of Appendix G.

7.2.3 | Lookahead Matching

Because the existing sellers can choose whether to use the lazy minting function, we conduct propensity score matching to mitigate self-selection bias between the two groups of existing sellers, but they may still differ on some unobserved characteristics driving their self-selection into different groups. Based on the lookahead matching method (Bapna et al. 2016; Hosanagar et al. 2014), we select the existing sellers who never used lazy minting in the window period but adopted it in the future as the control group. Given that these users eventually adopt the lazy minting function, they should share similar unobserved characteristics governing the decision to use lazy minting. Table G3 of Appendix G shows the regression results of the successful transaction analysis, which are similar to those reported in the main results. Our results are still robust.

7.3 | Long-Term Effects

We conduct further analyses to examine the long-term impact of the lazy minting policy by using the same user samples and doubling the window to 24 weeks after the event, and the results are shown in Table G4 of Appendix G. The results further reveal that in the long term, lazy minting still has the same significant effects on existing sellers' net sales performance.

7.4 | CSDID Method

We use the CSDID method (Callaway and Sant'Anna 2021), which proposes a group-time average treatment effect, to mitigate the heterogeneity of treatment effects. The results are shown in Table G5 of Appendix G. For all panels, the differences of all the dependent variables are not significant before the treatment (adopting lazy minting) and the effects after treatment are still consistent with our conclusions.

7.5 | Removal of NFTs Minted in Specific NFT Marketplaces

Different NFT marketplaces have different requirements for creators. Because there are no creator entry requirements in OpenSea and Rarible, these platforms host a diverse range of creators, from ordinary digital art creators to famous artists. Other platforms, however, have certain barriers for NFT minting. For example, Nifty Gateway has high requirements for creators, who must submit an application accompanied by a short video introduction stating the medium and long-term goals of their project. To avoid the requirements of different platforms affecting the results, we remove the NFTs minted on platforms with upfront creation barriers, only keeping the NFTs created in OpenSea and Rarible. The results are shown in Table G6. The adoption of the lazy minting policy still has the same effects on the sales performance of existing sellers with previous analyses.

8 | Discussion and Conclusions

In this study, leveraging a unique research context facilitated by OpenSea's implementation of lazy minting, we investigate whether and how adopting the function that allows optional upfront costs will be beneficial for existing sellers. While lazy minting offers a method to reduce ex ante costs, it concurrently leads to a diminution in IP protection strength. Consequently, sellers are confronted with the operational production decisions of selecting an appropriate minting approach that balances ex ante costs with ex post costs, contingent upon the type of their artwork (style-consistent vs. style-breaking). To investigate whether and how existing sellers should adopt lazy minting for better net sales performance, we conduct an empirical analysis based on the lazy minting implementation in OpenSea using a staggered DID method with PSM. To ensure the accuracy of the research design, we also carry out a series of robustness tests.

This study has three main findings. First, existing sellers enhance their net sales performance by adopting lazy minting. Second, this improvement in sales is attributed to the sellers' cost-adaptive IP protection behavior, which involves minting more NFTs with a greater proportion of style-consistent NFTs through lazy minting, while reserving regular minting for style-breaking NFTs. Third, the extent of this cost-adaptive IP protection behavior varies with the sellers' reputation, with those of higher reputation employing lazy minting more boldly. Moreover, evidence of this cost-adaptive IP protection behavior is observable on both the supply and demand sides.

Our study contributes to the existing literature in several dimensions. First, it unveils the impact and its mechanism of lazy minting—a novel NFT production method—on the net sales performance of existing sellers, thereby enriching the literature on digital transformation in operational decision changes (Angelopoulos et al. 2023; Heim et al. 2021). It enriches our understanding of the operational application of blockchain technology in two-sided markets, expanding our contemplation of its potential impact and implications (Babich and Hilary 2020). Furthermore, this research extends the application of TCE to the technological new phenomenon, which is traditionally utilized in analyzing corporate decisions such as outsourcing strategies and supply chain contract formulation (Benaroch et al. 2016). Here, TCE is expanded to blockchain-enabled two-sided markets and IP protection for digital creative artworks, enhancing our comprehension of transaction behavior changes within the NFT market through a cost analysis of various minting methods and artwork types. Additionally, our study augments the literature on NFT-based IP protection, which has largely focused on demonstrating the advantages of NFT-based IP protection or proposing specific technical protection solutions (Bamakan et al. 2022; Bonnet and Teuteberg 2023b). By analyzing the IP protection strength and demand for different types of NFTs, our research offers an economic perspective on NFT-based IP protection in the actual market. Lastly, the study makes a significant contribution to the literature on artists' style signatures (Sgourev and Althuizen 2014; Stamkou et al. 2018) by empirically proposing an AI algorithm-based style classification scheme and conceptually establishing a link between the style types of artworks and transaction costs.

Our research also holds practical significance. For sellers, it delineates a strategic operation approach to reduce transaction costs and augment profits. More importantly, it introduces a theoretical framework for sellers to devise personal strategies within a two-sided market amidst external uncertainties, particularly by evaluating and deciding from the perspectives of transaction costs and personal reputation. For platforms, our findings can provide guidance on whether the platform should adopt lazy minting. The initial intention of lazy minting was to lower market entry barriers and attract more new users,[25] thereby stimulating market innovation and improving overall liquidity. However, this study's findings suggest that reputable sellers would benefit much more than less reputable users, such as new sellers. It can be inferred that for artists newly entering the market who do not have much reputation, to ensure the IP protection strength of their works in the initial stages, they are more likely to utilize regular minting. Otherwise, the probability of their works being pirated or remaining unsold amidst numerous fraud lazy minted NFTs increases significantly.[26] This contradicts the original purpose of implementing lazy minting, which was primarily to help emerging artists reduce costs. This provides a plausible explanation for OpenSea's subsequent abandonment of this policy.

Furthermore, our conclusions effectively elucidate why Rarible continues to employ this policy. Given the differences in target audiences between the two platforms,[27] OpenSea focuses on all categories of NFTs and emphasizes being beginner-friendly, while Rarible primarily caters to mature artists.[28] Consequently, Rarible's primary users can strategically utilize lazy minting to reduce transaction costs and enhance trading performance. For OpenSea, the influx of vast quantities of pirated NFTs has adversely affected the entry of new artists they initially aimed to attract. The necessity to adopt regular minting and the increased market transaction friction have, in effect, raised entry barriers, resulting in a policy that primarily benefits existing sellers.

Like other empirical studies, our research has some limitations. First, our study only focuses on a specific segment of the users in the NFT market. Although the sales performance of existing sellers is the foundation of the platform economy and they can benefit through strategic utilization, we have to commit that this lazy minting policy may hold a negative impact from the market-level perspective. Second, we only consider the post-event periods of three months and six months. The longer-term impact of lazy minting on the users, as well as on the overall NFT ecosystem, is also worth more exploration. Third, we cannot conclusively determine how the overall creativity of the marketplace will evolve. As existing sellers who adopt lazy minting produce more style-consistent NFTs through this method, it remains an open question whether such cost-adaptive behavior will impact their individual creativity and, by extension, the creativity of the entire marketplace in the long term, which warrants further investigation. Moreover, even though we have matched and done the balanced test based on the observable pre-event covariates, there are still many unobservable factors affecting the lazy minting adoption, which may cause self-selection bias. The look-ahead matching strategy could alleviate this issue, but still not thoroughly.

NFTs are a crucial component of the Web 3.0 era, enabling the creation of unique digital assets and building the foundations of digital communities. However, as a novel form of digital transformation, the development of NFT marketplaces is not mature enough. In terms of market norms and trading systems, different marketplaces have made many attempts, such as different token standards, different royalty payments, and different minting methods. Our research focuses on the novel minting method in the NFT market, whose impact on users' operations and performance has not yet been fully understood. Despite the negative market-level implications, we find existing sellers can strategically adopt lazy minting based on considerations of transaction costs, artwork style, and personal reputation, thereby optimizing costs and enhancing net sales performance. This research represents an attempt to develop a conceptual and empirical understanding of the effects of optional IP protection methods on existing sellers in the two-sided markets and introduces transaction cost economics to explain the mechanism. We believe that the findings developed in this article could work as a basis for future blockchain studies in the field of operations management and provide the NFT markets with practice reference.

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[18] https://x.com/opensea/status/1486843204062236676?ref_src=twsrc%5Etfw.

[19] We also choose a longer window period of six months for robustness check.

[20] https://opensea.io/blog.

[21] For PSM, we restrict the matching to occur only within the common support region, potentially excluding some treatment group samples that do not find suitable matches. Also, we allow for the random selection of a control group sample when multiple controls have the same propensity score as a treatment unit. Therefore, if certain treatment samples fall outside the common support region, they will not be matched, leading to a reduced number of treatment samples.

[22] https://support.opensea.io/en/articles/8867014-who-pays-the-gas-fees-on-opensea.

[23] The results of parallel trend tests are shown in Appendix A. For brevity, we will not demonstrate it in the following sections.

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[25] https://101blockchains.com/lazy-minting/.

[26] https://www.vice.com/en/article/more-than-80-of-nfts-created-for-free-on-opensea-are-fraud-or-spam-company-says/.

[27] https://coinledger.io/learn/rarible-vs-opensea.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.